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# DATA SET DOCUMENTATION

## 1.Introduction

The effectiveness of a computer vision-based construction safety monitoring system is fundamentally dependent on the quality, diversity, and relevance of the dataset used for training and evaluation. In safety-critical applications such as construction environments, datasets must not only support accurate object detection but also enable reliable identification of unsafe conditions.

In this project, a **hybrid dataset approach** was adopted, combining a large-scale publicly available dataset with a custom-collected dataset. This strategy ensures both **high-volume learning capability** and **real-world adaptability**, while also fulfilling the requirement to demonstrate independent data collection and curation as specified in the assignment guidelines

## 2.Dataset Collection Strategy

The dataset was constructed using a **two-component strategy**:

1. **Primary Dataset (Public PPE Dataset)**
2. **Custom Dataset (Self-collected and annotated images)**

The primary dataset provides large-scale annotated data necessary for deep learning models, while the custom dataset addresses specific weaknesses identified during initial model evaluation.

This combined approach enhances both **generalization performance** and **edge-case handling**, particularly for underrepresented scenarios such as non-standard PPE variations.

## 3.Primary  Dataset Description

The primary dataset was sourced from the **Roboflow PPE Dataset**, which contains annotated images representing construction and industrial environments .

The dataset includes **43,999 images** annotated in YOLO format, making it directly compatible with the selected object detection framework.

### Dataset Split

- Training Set: 30,765 images
- Validation Set: 8,814 images
- Test Set: 4,423 images

Data validation confirmed that the dataset is clean, with no missing images, missing labels, or corrupted annotations.

## 4.Class Definitions

The dataset consists of **14 classes**, categorized into three groups:

### 4.1 PPE Detection Classes

- Hardhat
- Safety Vest
- Gloves
- Goggles
- Mask

### 4.2 Safety Violation Classes

- NO-Hardhat
- NO-Safety Vest
- NO-Gloves
- NO-Goggles
- NO-Mask

### 4.3 Contextual Classes

- Person
- Safety Cone
- Ladder
- Fall-Detected

This class structure enables the system to detect both **presence of safety equipment** and **violations directly**, supporting higher-level reasoning required by the task

## 5.Dataset Distribution Analysis

The training dataset contains approximately **76,398 bounding box annotations** .

Analysis of class distribution shows:

- **High-frequency classes**: Hardhat, Safety Cone
- **Medium-frequency classes**: Safety Vest, NO-Gloves
- **Low-frequency classes**: Ladder, Mask

This indicates the presence of **class imbalance**, which is typical in real-world datasets. However, critical safety-related classes are sufficiently represented to ensure meaningful learning.

## 6.Dataset Characteristics

The dataset includes a wide range of real-world variations, ensuring strong generalization capability.

### 6.1 Environment Diversity

- Indoor construction sites
- Outdoor work zones
- Warehouses
- Scaffolding areas

### 6.2 Lighting Conditions

- Daylight
- Shadow
- Artificial lighting
- Low-light environments

### 6.3 Scene Complexity

- Multiple workers per frame
- Occlusions
- Different viewing angles
- Varying distances

These variations align with assignment requirements for dataset diversity .

## 7.Custom Dataset Creation

To improve model performance and satisfy assignment requirements, a **custom dataset of 125 images** was created.

### 7.1 Motivation

During initial testing, it was observed that the model struggled to detect:

- **blue-colored safety vests**
- non-standard PPE variations

This issue occurs because most public datasets are biased toward:

- yellow safety vests
- orange safety gear

### 7.2 Data Collection Sources

Custom images were collected from:

- Pexels
- Unsplash

These platforms provide **royalty-free images**, ensuring:

- legal compliance
- no copyright violations
- ethical dataset usage

### 7.3 Data Selection Criteria

Images were selected based on:

- relevance to construction or industrial environments
- presence of workers
- visibility of PPE

- inclusion of blue safety vests

Irrelevant images were excluded to maintain dataset quality.

### 7.4 Annotation Process

- Images were manually annotated using bounding boxes
- Classes included:
  - Safety Vest
  - Hardhat
  - Persons
- Annotation was performed using Robow Flow
- YOLO format was maintained for consistency

### 7.5 Dataset Size

- Total images: **125**

### 7.6 Impact of Custom Dataset

The custom dataset improves:

- detection of non-standard PPE colors
- reduction of false negatives
- real-world performance

It also demonstrates:

- data curation skills
- problem-driven dataset enhancement

## 8. Data Integrity and Security

Strict validation and ethical standards were maintained throughout dataset preparation.

Measures taken include:

- Use of only licensed or royalty-free images
- No inclusion of personal or sensitive data
- Verification of all annotations

Dataset validation confirmed:

- No missing labels
- No corrupted files

- Consistent annotation format

## 9. Combined Dataset Strategy

The final dataset used for training is a **merged dataset** consisting of:

- Public dataset (~44,000 images)
- Custom dataset (125 images)

This hybrid approach enables:

- Strong baseline performance
- Improved handling of edge cases
- Better adaptability to real-world scenarios

## 10. Limitations

Despite improvements, some limitations remain:

- class imbalance across categories
- limited size of custom dataset
- difficulty in detecting PPE under heavy occlusion

These limitations are acknowledged and considered in system evaluation.

## 11. Summary

The dataset used in this project combines:

- large-scale annotated data
- targeted custom data
- diverse real-world conditions

This approach enables the model to achieve:

- accurate PPE detection
- reliable safety violation identification
- improved generalization

# 12.Appendix

Main Dataset link:

Personal Protective Equipment - Combined Model Object Detection Model by Roboflow ...

Custom Dataset Public Project Link: ppe_ext_data Object Detection Dataset by AIML

Count Images and Labels in Each Split

This cell counts how many image files and label files are present in each dataset split, helping confirm completeness and detect missing files.

```python
from glob import glob
import os

IMAGE_EXTS = ["*.jpg", "*.jpeg", "*.png", "*.bmp", "*.webp", "*.JPG", "*.JPEG", "*.PNG"]

def count_images(folder):
    total = 0
    for ext in IMAGE_EXTS:
        total += len(glob(os.path.join(folder, ext)))
    return total

def count_labels(folder):
    return len(glob(os.path.join(folder, "*.txt")))

print("Train images:", count_images(train_path))
print("Train labels:", count_labels(train_labels_path))
print("Val images  :", count_images(val_path))
print("Val labels  :", count_labels(val_labels_path))
print("Test images :", count_images(test_path))
print("Test labels :", count_labels(test_labels_path))
```

```
Train images: 30765
Train labels: 30765
Val images  : 8814
Val labels  : 8814
Test images : 4423
Test labels : 4423
```

*Image1:Main dataset overview*

```
===== TRAIN =====
Image files          : 30765
Label files          : 30765
Missing labels       : 0
Missing images       : 0
Sample missing label: []
Sample missing image: []

===== VALID =====
Image files          : 8814
Label files          : 8814
Missing labels       : 0
Missing images       : 0
Sample missing label: []
Sample missing image: []

===== TEST =====
Image files          : 4423
Label files          : 4423
Missing labels       : 0
Missing images       : 0
Sample missing label: []
Sample missing image: []
```

*Image2: Image and label distribution on splits*

## Standardize Class Name Format

This cell converts the class definitions into a consistent list format, making them easier to use in charts, summaries, and label interpretation.

```python
if isinstance(names, dict):
    class_names = [names[i] for i in sorted(names.keys())]
else:
    class_names = names

print("Ordered class list:")
for i, name in enumerate(class_names):
    print(i, "->", name)
```

```
Ordered class list:
0 -> Fall-Detected
1 -> Gloves
2 -> Goggles
3 -> Hardhat
4 -> Ladder
5 -> Mask
6 -> NO-Gloves
7 -> NO-Goggles
8 -> NO-Hardhat
9 -> NO-Mask
10 -> NO-Safety Vest
11 -> Person
12 -> Safety Cone
13 -> Safety Vest
```

*Image3: class and class id distribution*

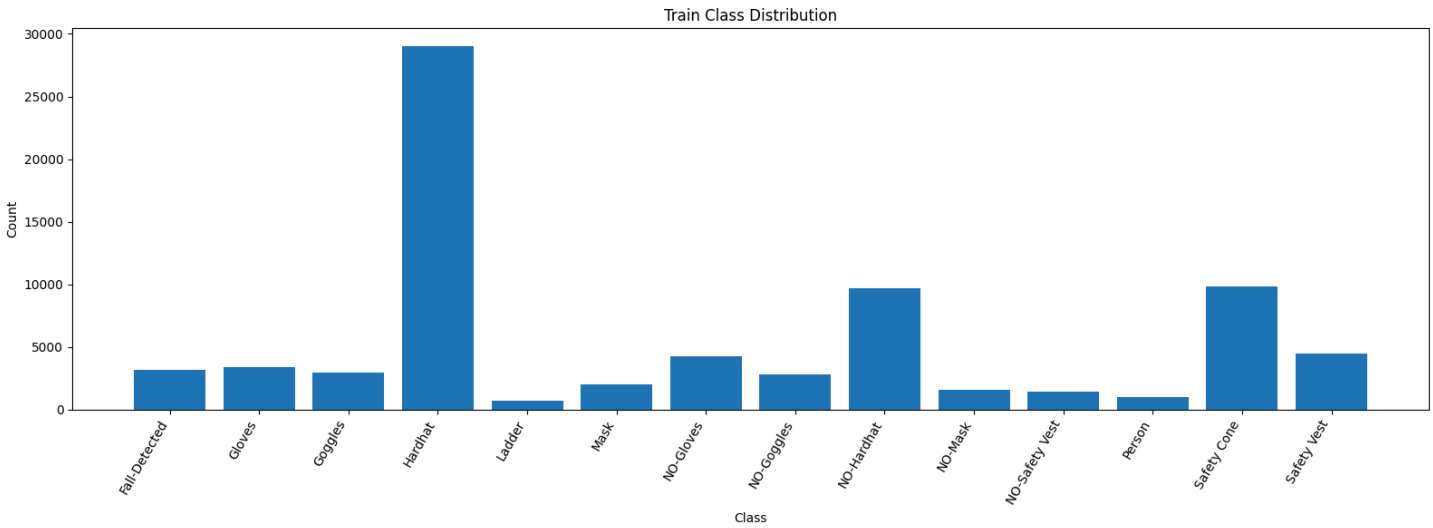


*Image4: Train class distribution*

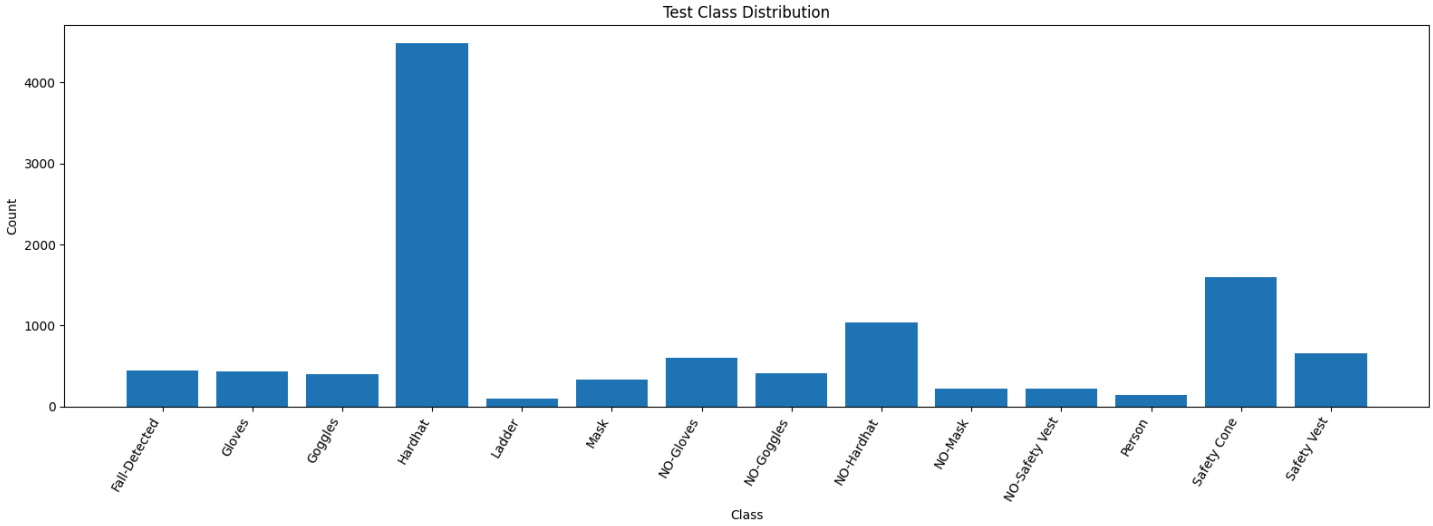


*Image5: Test class distribution*

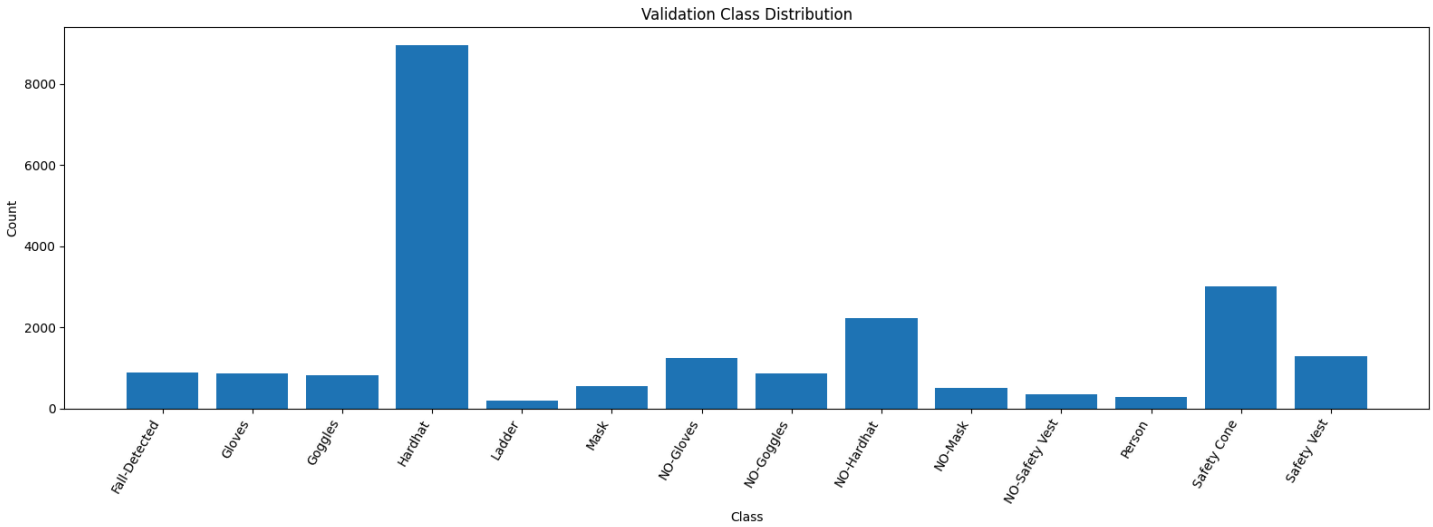


*Image6: Validation class distribution*

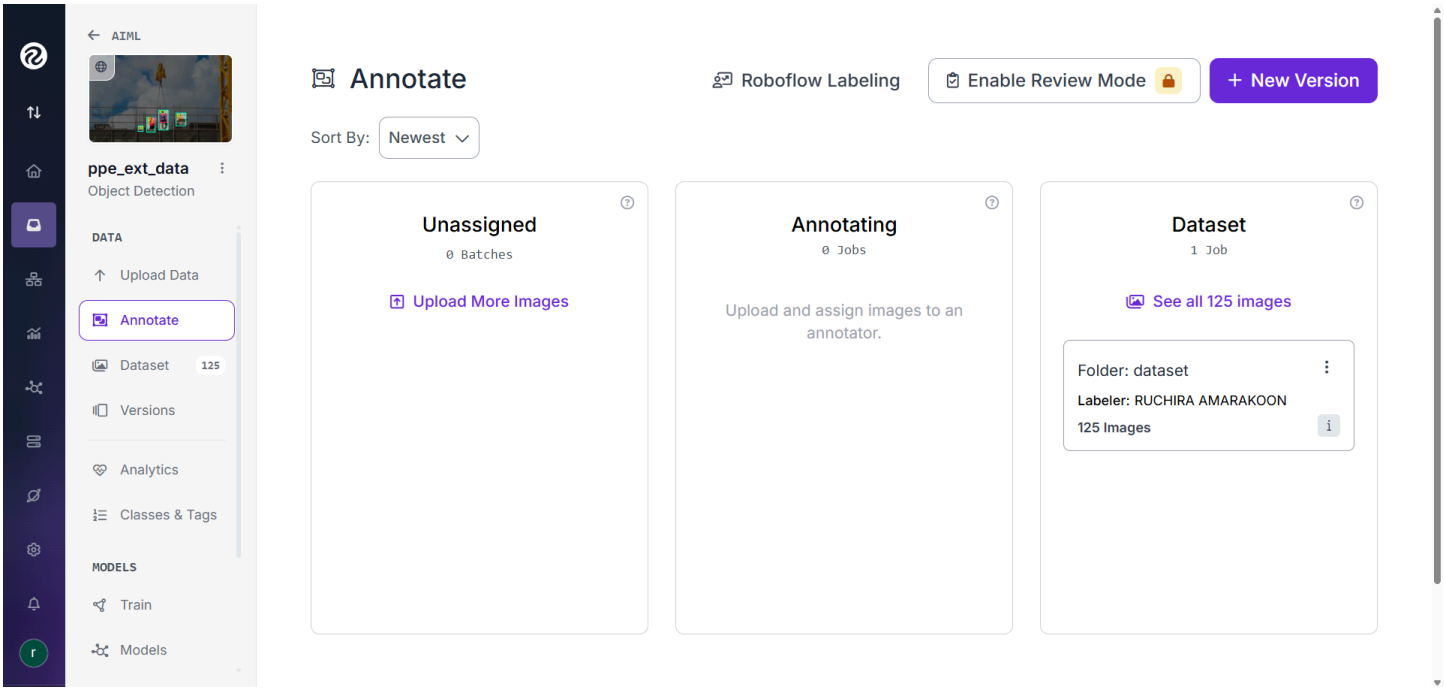


*Image7:Robow flow data annotation(custom dataset)*

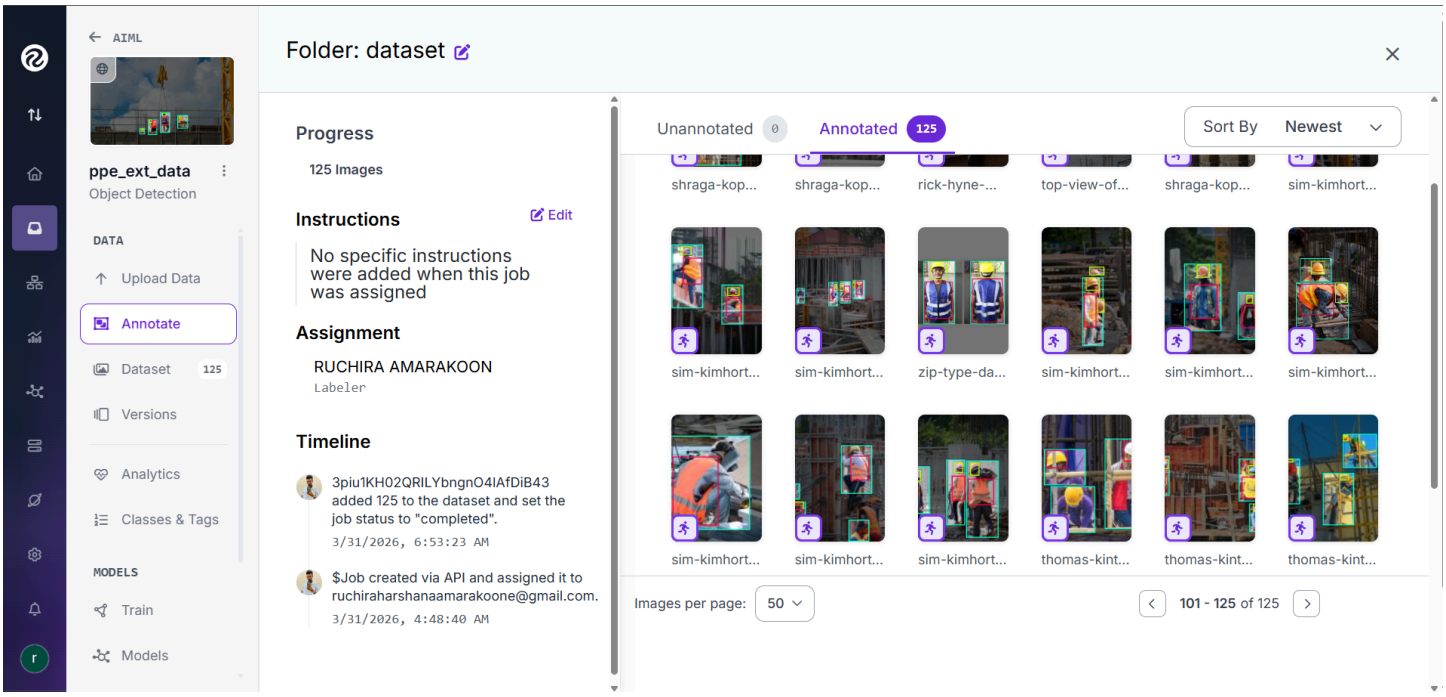


*Image8:Robow flow data annotation(custom dataset)*

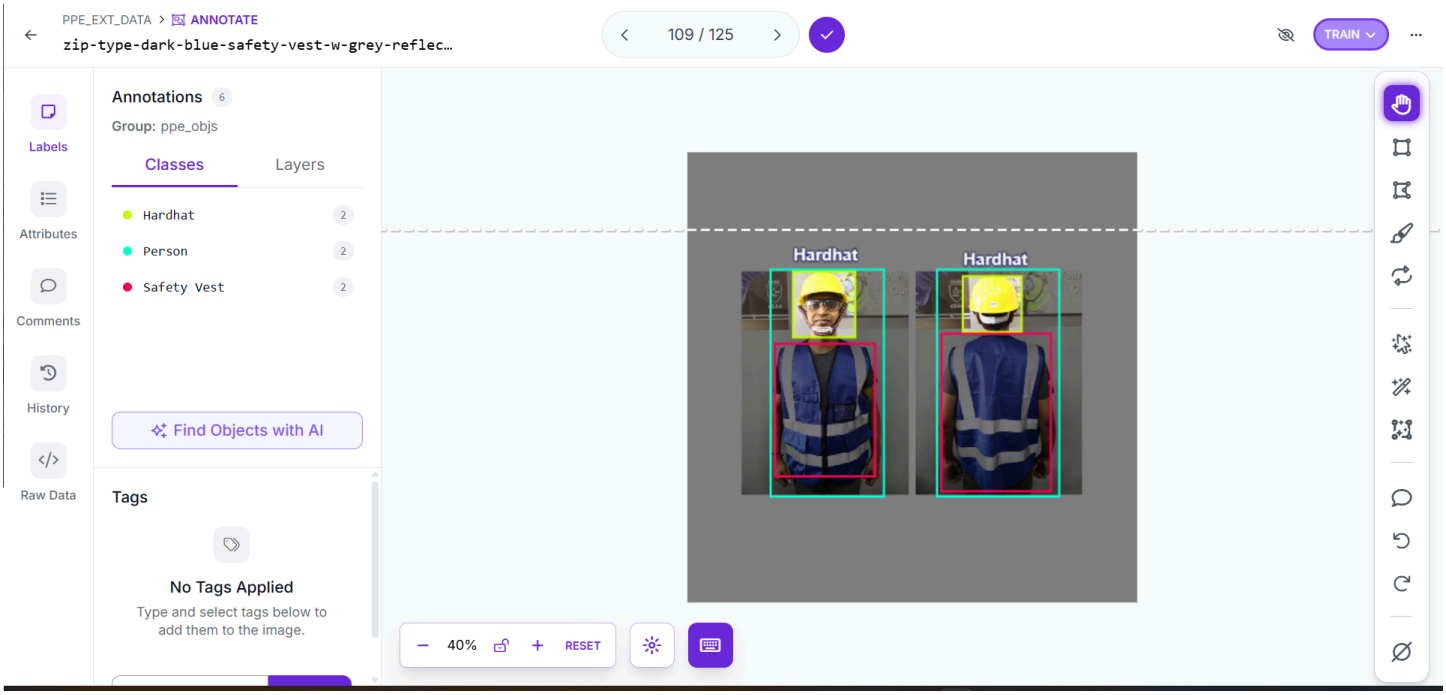


*Image8:Sample blue vest annotation (custom dataset)*